Advanced Topics in Computer Graphics II
Geometry Processing

Registration

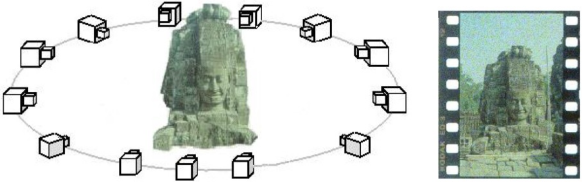
January 13, 2025

3D Photography consists of 3 steps:

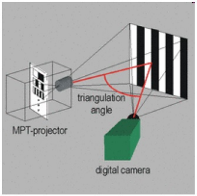
1. Acquisition phase
   $\rightarrow$ Gathering of raw data, generation of initial geometry, in most cases point clouds or contour lines
2. Registration phase
   $\rightarrow$ Alignment of individual scan data sets in a common coordinate system
3. Reconstruction phase
   $\rightarrow$ Conversion of the point cloud into a standard representation like polygonal meshes dealing with outliers, holes, and other acquisition artifacts

Acquisition phase

There are several technologies to capture 3D data, e.g.:



(a) Image capturing



(b) Distance measurement

(c) Hand-held scanning

Goal: find pairs of corresponding points between shapes represented as point sets to

- organize
- reconstruct
- synthesize

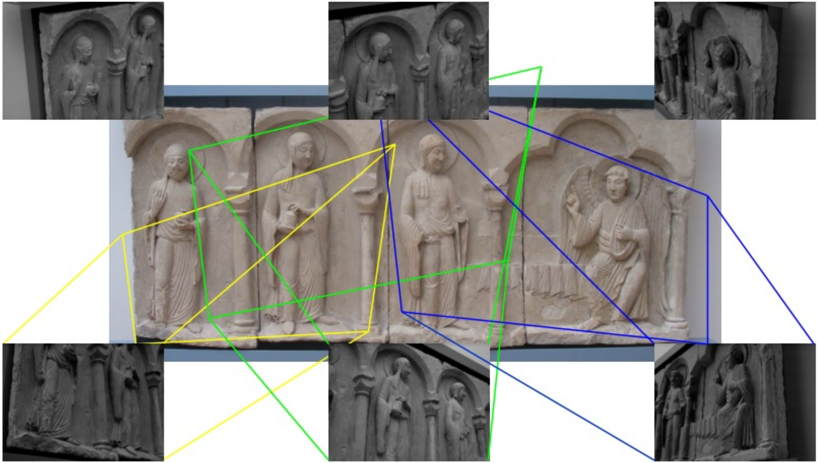
a broader class of shapes.

- **rigid** registration: find a map that preserves the distance between every pair of points
- **no-rigid** registration: find a map that does not preserve the distance between every pair of points but preserve the topology of the shapes

Probabilistic setting: Model uncertainty

- **Coherent point drift** (CPD) is a collection of different registration algorithms based on a Gaussian Mixture Model (GMM).
- performance, scalability to large data sets and stability/robustness of the correspondence estimation due to soft-matching

Only a few objects can be captured completely. Several range images are needed to capture the complete object



More precisely, the following task has to be solved:

- Every range-map describes only one part of the object surface (viewport)
- Due to occlusions, it might be impossible to capture the whole object
- Have to deal with partial views of the object in local coordinate systems

Registration Task: Transformation into a common coordinate system!

Rigid Registration: Point Cloud ICP

Given: Two point clouds

$$\mathcal{P} = \{\boldsymbol{p}_1, \ldots, \boldsymbol{p}_m\} \quad , \quad \mathcal{Q} = \{\boldsymbol{q}_1, \ldots, \boldsymbol{q}_n\}$$

Task: (Registration of point clouds)

Find a rigid transformation

$$\boldsymbol{T} = (\boldsymbol{R}, \boldsymbol{t})$$

with $\boldsymbol{R} \in \mathbb{R}^{3\times 3}, \boldsymbol{t} \in \mathbb{R}^3, \boldsymbol{R}^T \boldsymbol{R} = \boldsymbol{I}$ such that some matching cost/energy is minimized.

- Energy (point-to-point):

$$E = E(\boldsymbol{R}, \boldsymbol{t}) = \sum_{\boldsymbol{p}_i \in \mathcal{P}} \sum_{\boldsymbol{q}_j \in \mathcal{Q}} w_{ij} \|\boldsymbol{p}_i - (\boldsymbol{R}\,\boldsymbol{q}_j + \boldsymbol{t})\|^2$$

  where w_{ij} is a weight indicating the correspondence between $\boldsymbol{p}_i$ and $\boldsymbol{q}_j$
- Other energy formulations possible, e.g. point-to-plane, NDT, ...

Energy (point-to-point):

$$E = E(\boldsymbol{R}, \boldsymbol{t}) = \sum_{\boldsymbol{p}_i \in \mathcal{P}} \sum_{\boldsymbol{q}_j \in \mathcal{Q}} w_{ij} \|\boldsymbol{p}_i - (\boldsymbol{R}\,\boldsymbol{q}_j + \boldsymbol{t})\|^2$$

where w_{ij} is a weight indicating the correspondence between $\boldsymbol{p}_i$ and $\boldsymbol{q}_j$

Problem: All of these energies are non-convex!

Theorem (Murty and Kabadi (1987)[1])

If an optimization problem is non-convex, it is NP-hard.

$\rightarrow$ So in general, there is no chance to efficiently calculate the global optimum (for more details see Manyem and Ugon (2012) and related papers).

Idea: Use heuristics to approximate the global optimum!

[1] K. G. Murty and S. N. Kabadi. "Some NP-complete Problems in Quadratic and Nonlinear Programming". In: *Mathematical programming* 39.2 (1987), pp. 117–129.

One famous registration algorithm is the **I**terative-**C**losest-**P**oint algorithm.

Heuristic: Assume ...

- there exists a set of correspondences such that the optimal alignment of this set is also optimal for the whole problem
- the weights are binary: $w_{ij} \in \{0, 1\}$
- one can "decouple correspondence optimization from transformation optimization"

Idea: Similar to k-means, iteratively do ...

- ... first optimize the correspondences and keep $\boldsymbol{T} = (\boldsymbol{R}, \boldsymbol{t})$ fixed ...
- ... then optimize $\boldsymbol{T} = (\boldsymbol{R}, \boldsymbol{t})$ and keep the correspondences fixed.

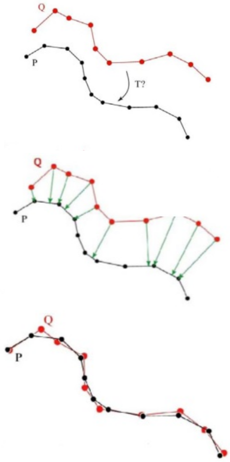
$\rightarrow$ Gradient descent technique of the energy

$$E(\boldsymbol{R}, \boldsymbol{t}) = \sum_{i=1}^{n} \left\| \boldsymbol{p}_{c(i)} - (\boldsymbol{R}\,\boldsymbol{q}_i + \boldsymbol{t}) \right\|^2$$

Algorithm: ICP[2]

1. Start from an approximate registration
2. Repeat ...
   - Identify corresponding points $(\boldsymbol{q}_i, \boldsymbol{p}_{c(i)})$
   - Compute and apply the optimal rigid transformation $\boldsymbol{T} = (\boldsymbol{R}, \boldsymbol{t})$
3. ... until registration error (point-to-point distance) is small

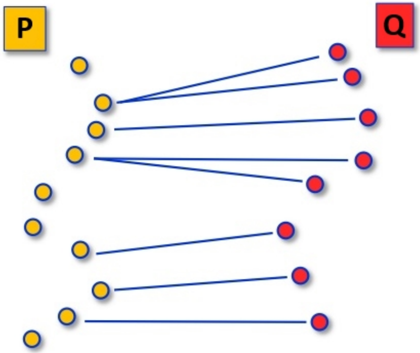
$$E(\boldsymbol{R}, \boldsymbol{t}) = \sum_{i=1}^{n} \left\| \boldsymbol{p}_{c(i)} - (\boldsymbol{R}\,\boldsymbol{q}_i + \boldsymbol{t}) \right\|^2 < \delta_{threshold}$$



[2] P. Besl and N. D. McKay. "A Method for Registration of 3-D Shapes". In: *Pattern Analysis and Machine Intelligence, IEEE Transactions on* 14.2 (1992), pp. 239–256.

Problem: How to compute correspondences?

Heuristic: Choose the closest point of the other point cloud



Problem: How to compute an optimal transformation $\boldsymbol{T} = (\boldsymbol{R}, \boldsymbol{t})$ out of the correspondences?

Theorem (Centers of Mass)

If $\boldsymbol{T} = (\boldsymbol{R}, \boldsymbol{t})$ is an optimal transformation, then the correspondence points $\{\boldsymbol{p}_{c(i)}\}$ and $\{\boldsymbol{R}\,\boldsymbol{q}_i + \boldsymbol{t}\}$ have the same centers of mass, i.e.

$$\overline{\boldsymbol{p}} = \frac{1}{n}\sum_{i=1}^{n} \boldsymbol{p}_{c(i)} = \frac{1}{n}\sum_{i=1}^{n} (\boldsymbol{R}\,\boldsymbol{q}_i + \boldsymbol{t}) = \boldsymbol{R}\,\overline{\boldsymbol{q}} + \boldsymbol{t}$$

Lemma (Least Squares Distance Minimizer)

Let $\{\boldsymbol{x}_i\}$ be a set of points and $\overline{\boldsymbol{x}} = \frac{1}{n}\sum_{i=1}^{n} \boldsymbol{x}_i$ be their center of mass. Then $\overline{\boldsymbol{x}}$ minimizes the following energy

$$E(\boldsymbol{x}) = \sum_{i=1}^{n} \|\boldsymbol{x} - \boldsymbol{x}_i\|^2$$

This can easily be seen by rewriting the energy and calculating its gradient:

$$
\begin{aligned}
E(\boldsymbol{x}) &= \sum_{i=1}^{n} \|\boldsymbol{x} - \boldsymbol{x}_i\|^2 \\
&= \sum_{i=1}^{n} (\boldsymbol{x} - \boldsymbol{x}_i)^T (\boldsymbol{x} - \boldsymbol{x}_i) \\
&= \sum_{i=1}^{n} \left(\boldsymbol{x}^T \boldsymbol{x} - 2\, \boldsymbol{x}^T \boldsymbol{x}_i + \boldsymbol{x}_i^T \boldsymbol{x}_i\right)
\end{aligned}
$$

Therefore, we get

$$
\nabla E(\boldsymbol{x}) = \sum_{i=1}^{n} (2\, \boldsymbol{x} - 2\, \boldsymbol{x}_i)
$$

Setting the gradient to zero gives the desired result:

$$
\sum_{i=1}^{n} 2\, \boldsymbol{x} = \sum_{i=1}^{n} 2\, \boldsymbol{x}_i \quad \Leftrightarrow \quad n\, \boldsymbol{x} = \sum_{i=1}^{n} \boldsymbol{x}_i \quad \Leftrightarrow \quad \boldsymbol{x} = \frac{1}{n} \sum_{i=1}^{n} \boldsymbol{x}_i
$$

If we now consider the difference vectors $\boldsymbol{p}_{c(i)} - \boldsymbol{R}\,\boldsymbol{q}_i$ for any given rotation $\boldsymbol{R}$, then according to the lemma the vector minimizing the squared distances to these difference vectors is

$$\boldsymbol{t} = \frac{1}{n}\sum_{i=1}^{n}\bigl(\boldsymbol{p}_{c(i)} - \boldsymbol{R}\,\boldsymbol{q}_i\bigr)$$

In other words, this vector minimizes the translational registration error. Written in terms of the centers of mass we get

$$\boldsymbol{t} = \left(\frac{1}{n}\sum_{i=1}^{n}\boldsymbol{p}_{c(i)}\right) - \boldsymbol{R}\left(\frac{1}{n}\sum_{i=1}^{n}\boldsymbol{q}_i\right) = \overline{\boldsymbol{p}} - \boldsymbol{R}\,\overline{\boldsymbol{q}}$$

which directly leads to the claimed equation in the theorem:

$$\overline{\boldsymbol{p}} = \boldsymbol{R}\,\overline{\boldsymbol{q}} + \boldsymbol{t}$$

So first, an optimal rotation can be computed and then the translation is given as

$$\boldsymbol{t} = \overline{\boldsymbol{p}} - \boldsymbol{R}\,\overline{\boldsymbol{q}}$$

Defining

$$\boldsymbol{p}_i^{'} = \boldsymbol{p}_i - \overline{\boldsymbol{p}} \quad , \quad \boldsymbol{q}_i^{'} = \boldsymbol{q}_i - \overline{\boldsymbol{q}}$$

and using the above result, the error function can be rewritten and reduced to

$$\begin{aligned}
E(\boldsymbol{R}, \boldsymbol{t}) &= \sum_{i=1}^{n} \left\| \boldsymbol{p}_{c(i)} - (\boldsymbol{R}\,\boldsymbol{q}_i + \boldsymbol{t}) \right\|^2 \\
&= \sum_{i=1}^{n} \left\| \boldsymbol{p}_{c(i)}^{'} + \overline{\boldsymbol{p}} - \left(\boldsymbol{R}\,(\boldsymbol{q}_i^{'} + \overline{\boldsymbol{q}}) + \boldsymbol{t} \right) \right\|^2 \\
&= \sum_{i=1}^{n} \left\| \boldsymbol{p}_{c(i)}^{'} - \boldsymbol{R}\,\boldsymbol{q}_i^{'} + \underbrace{\overline{\boldsymbol{p}} - \boldsymbol{R}\,\overline{\boldsymbol{q}} - \boldsymbol{t}}_{=0} \right\|^2 \\
&= \sum_{i=1}^{n} \left\| \boldsymbol{p}_{c(i)}^{'} - \boldsymbol{R}\,\boldsymbol{q}_i^{'} \right\|^2
\end{aligned}$$

This leads to

$$
\begin{aligned}
&\arg\min_{\boldsymbol{R}} \sum_{i=1}^{n} \left\| \boldsymbol{p}_{c(i)}^{'} - \boldsymbol{R}\,\boldsymbol{q}_i^{'} \right\|^2 \\
=&\arg\min_{\boldsymbol{R}} \sum_{i=1}^{n} \left(\boldsymbol{p}_{c(i)}^{'} - \boldsymbol{R}\,\boldsymbol{q}_i^{'} \right)^T \left(\boldsymbol{p}_{c(i)}^{'} - \boldsymbol{R}\,\boldsymbol{q}_i^{'} \right) \\
=&\arg\min_{\boldsymbol{R}} \sum_{i=1}^{n} \boldsymbol{p}_{c(i)}^{'T}\,\boldsymbol{p}_{c(i)}^{'} - \boldsymbol{p}_{c(i)}^{'T}\,\boldsymbol{R}\,\boldsymbol{q}_i^{'} - \boldsymbol{q}_i^{'T}\,\boldsymbol{R}^T\,\boldsymbol{p}_{c(i)}^{'} + \boldsymbol{q}_i^{'T}\,\underbrace{\boldsymbol{R}^T\,\boldsymbol{R}}_{=\boldsymbol{I}}\,\boldsymbol{q}_i^{'} \\
=&\arg\min_{\boldsymbol{R}} \sum_{i=1}^{n} -\boldsymbol{p}_{c(i)}^{'T}\,\boldsymbol{R}\boldsymbol{q}_i^{'} - \underbrace{\left(\boldsymbol{q}_i^{'T}\,\boldsymbol{R}^T\,\boldsymbol{p}_{c(i)}^{'} \right)^T}_{\text{this is a scalar}} \\
=&\arg\min_{\boldsymbol{R}} \sum_{i=1}^{n} -2 \cdot \boldsymbol{p}_{c(i)}^{'T}\,\boldsymbol{R}\,\boldsymbol{q}_i^{'} \\
=&\arg\max_{\boldsymbol{R}} \sum_{i=1}^{n} \boldsymbol{p}_{c(i)}^{'T}\,\boldsymbol{R}\,\boldsymbol{q}_i^{'}
\end{aligned}
$$

Now, this error can further be rewritten as the trace (sum of diagonal entries) of a matrix:

$$\begin{aligned}
&\arg\max_{\boldsymbol{R}} \sum_{i=1}^{n} \boldsymbol{p}_{c(i)}^{'T}\, \boldsymbol{R}\, \boldsymbol{q}_i^{'} \\
&= \arg\max_{\boldsymbol{R}} \operatorname{tr}\left(\sum_{i=1}^{n} \boldsymbol{p}_{c(i)}^{'T}\, \boldsymbol{R}\, \boldsymbol{q}_i^{'} \right) && \text{since a scalar is a 1D matrix} \\
&= \arg\max_{\boldsymbol{R}} \operatorname{tr}\left(\sum_{i=1}^{n} \boldsymbol{R}\, \boldsymbol{q}_i^{'}\, \boldsymbol{p}_{c(i)}^{'T} \right) && \text{since } \operatorname{tr}(A^T B) = \operatorname{tr}(B A^T) \\
&= \arg\max_{\boldsymbol{R}} \operatorname{tr}\left(\boldsymbol{R} \sum_{i=1}^{n} \boldsymbol{q}_i^{'}\, \boldsymbol{p}_{c(i)}^{'T} \right)
\end{aligned}$$

Now define

$$\boldsymbol{H} := \sum_{i=1}^{n} \boldsymbol{q}_i^{'}\, \boldsymbol{p}_{c(i)}^{'T}$$

This matrix measures the "covariance between the correspondences".

So the rotation can be found by

$$\arg\max_{\boldsymbol{R}} \operatorname{tr}(\boldsymbol{R}\,\boldsymbol{H})$$

Theorem (Matrix Trace)

If $\boldsymbol{M}$ is a symmetric positive semi-definite matrix (all eigenvalues are non-negative), then for any orthonormal matrix $\boldsymbol{B}$

$$\operatorname{tr}(\boldsymbol{M}) \geq \operatorname{tr}(\boldsymbol{B}\,\boldsymbol{M})$$

$\rightarrow$ So find a rotation matrix $\boldsymbol{R}$ such that $\boldsymbol{R}\,\boldsymbol{H}$ is positive semi-definite and the theorem will guarantee that $\boldsymbol{R}$ is the solution of our problem!

This can easily be done by a Singular Value Decomposition:

$$\boldsymbol{H} = \boldsymbol{U}\,\boldsymbol{\Sigma}\,\boldsymbol{V}^T$$

Now, define

$$\boldsymbol{R} := \boldsymbol{V}\,\boldsymbol{U}^T$$

Since $\boldsymbol{U}$ and $\boldsymbol{V}$ are orthonormal matrices, $\boldsymbol{R}$ is also orthonormal and thus a rotation.

Furthermore, the following holds:

$$\boldsymbol{R}\,\boldsymbol{H} = \boldsymbol{V}\,\boldsymbol{U}^T\,\boldsymbol{U}\,\boldsymbol{\Sigma}\,\boldsymbol{V}^T = \boldsymbol{V}\,\boldsymbol{\Sigma}\,\boldsymbol{V}^T$$

So we see that this is an eigenvalue decomposition of $\boldsymbol{R}\,\boldsymbol{H}$. Therefore, $\boldsymbol{R}\,\boldsymbol{H}$ is symmetric and positive semi-definite.

Note: To have a unique solution, $\boldsymbol{H}$ needs to have full rank! Otherwise $\boldsymbol{R}$ will not be unique since at least one of its singular values is zero. This means that the data are co-planar, so registration may fail.

Theorem (Iterative Closest Point algorithm)

The ICP algorithm converges to a local minimum of the error function (point-to-point distance)

$$E(\boldsymbol{R}, \boldsymbol{t}) = \sum_{i=1}^{n} \left\| \boldsymbol{p}_{c(i)} - (\boldsymbol{R}\,\boldsymbol{q}_i + \boldsymbol{t}) \right\|^2$$

Proof.

The idea is the following:

1. The minimization of E cannot increase the sum of squared distances between corresponding points
2. The new correspondences in the next iteration does also not increase this distance

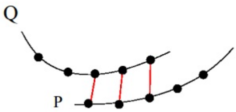
□

(Detailed proofs of the mentioned theorems can be found in Horn et al. (1988)[3]).

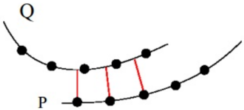
[3] B. K. Horn et al. "Closed-form Solution of Absolute Orientation using Orthonormal Matrices". In: *JOSA A* 5.7 (1988), pp. 1127–1135.

Limitations:

1. Non-convex optimization $\rightarrow$ only local minimum guaranteed
2. Initial alignment crucial!
3. Problem of finding correspondences
   - Standard solution: closest neighbor + some heuristics
   - Chen and Medioni (1991) : Minimize point-to-plane distance

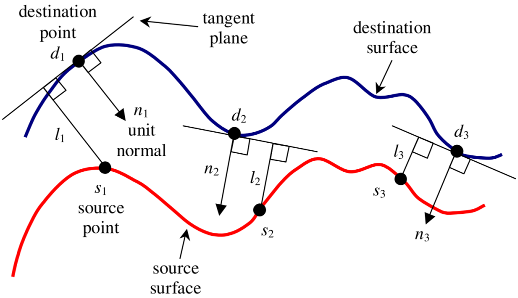


match closest neighbor in space



project point onto plane

The following variant[4] of the ICP uses the distances to the tangent planes as the objective function (normal estimation is discussed later):

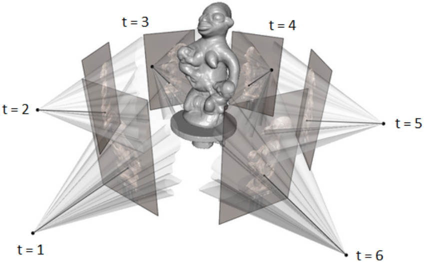


The error is defined as the point-to-plane error:

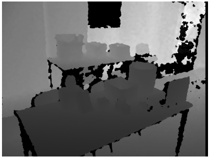
$$E = E(\boldsymbol{R}, \boldsymbol{t}) = \sum_{i=1}^{n} \left\langle \boldsymbol{p}_{c(i)} - (\boldsymbol{R}\,\boldsymbol{q}_i + \boldsymbol{t}) \,\middle|\, \boldsymbol{n}_{c(i)} \right\rangle^2$$

[4] Y. Chen and G. Medioni. "Object Modeling by Registration of Multiple Range Images". In: *Robotics and Automation, 1991. Proceedings., 1991 IEEE International Conference on*. IEEE. 1991, pp. 2724–2729.

RGB-D ICP



(a) RGB image $\boldsymbol{I}_{rgb}$



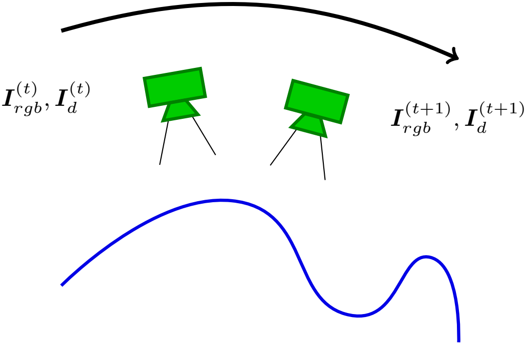
(b) Depth image $\boldsymbol{I}_d$

Given: Incremental stream of sensor data by the Microsoft Kinect sensor

$$\boldsymbol{\mathcal{I}}_{rgb} = \{\boldsymbol{I}_{rgb}^{(t)} \mid t = 1, 2, \ldots\}$$

$$\boldsymbol{\mathcal{I}}_{d} = \{\boldsymbol{I}_{d}^{(t)} \mid t = 1, 2, \ldots\}$$

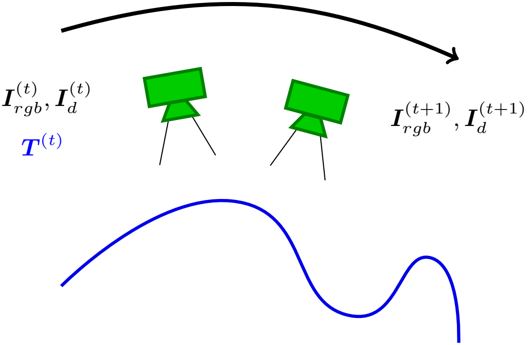
Problem: RGB-D images are given in local coordinates



Task: Transform the data to global coordinates, i.e. find the rigid transformation

$$\boldsymbol{T}^{(t+1)} = \begin{bmatrix} \boldsymbol{R}^{(t+1)} & \boldsymbol{t}^{(t+1)} \\ \boldsymbol{0}^T & 1 \end{bmatrix} \in \mathbb{R}^{4\times 4}$$

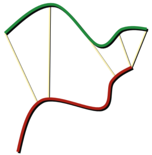
Task: Transform the data to global coordinates, i.e. find the rigid transformation $\boldsymbol{T}^{(t+1)}$

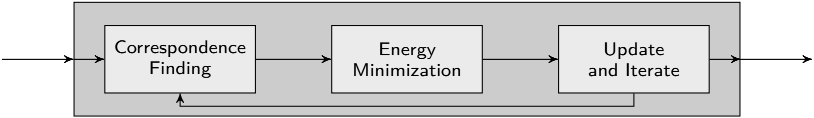


Idea: Since the data are captured incrementally, we can use the global transformation from the previous frame $\boldsymbol{T}^{(t)}$ and only compute the relative transformation between the two frames $\boldsymbol{\Delta T}$. Then we can infer the current global transformation using

$$\boldsymbol{T}^{(t+1)} = \boldsymbol{T}^{(t)} \cdot \boldsymbol{\Delta T}$$

Question: How to compute $\boldsymbol{\Delta T}$?

KinectFusion ICP (Newcombe et al., 2011)	Photometric ICP (Steinbrücker et al., 2011)
	

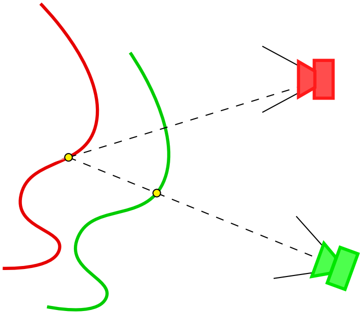


1. Compute correspondences between source points $\boldsymbol{I}_d^{(t+1)}$ and target points $\boldsymbol{I}_d^{(t)}$
   - To indicate the corresponding frames at time $t+1$ and t in image-, sensor- and model coordinates $\boldsymbol{p}, \boldsymbol{v}$ and $\hat{\boldsymbol{v}}$, let

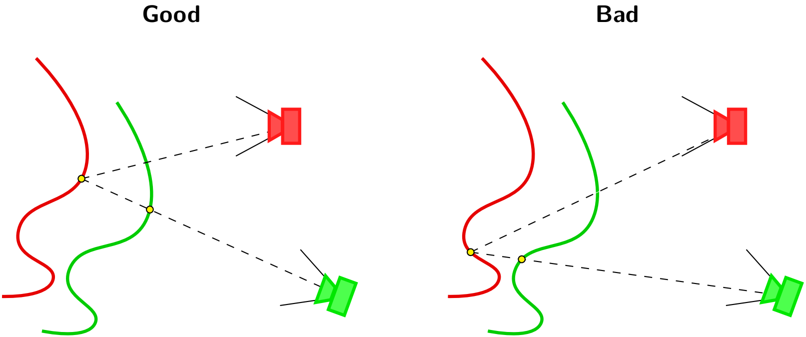
$$\hat{\boldsymbol{v}}_i^{(t)} := \hat{\boldsymbol{v}}^{(t)}(\boldsymbol{p}_i)$$
$$\boldsymbol{v}_j^{(t+1)} := \boldsymbol{\Delta T} \cdot \boldsymbol{v}^{(t+1)}(\boldsymbol{p}_j)$$

   - Each source 3D point $\boldsymbol{v}_i^{(t+1)}$ projects to pixel $\boldsymbol{p}_{c(i)}$ in target camera
   - Use the 3D point $\hat{\boldsymbol{v}}_{c(i)}^{(t)}$ of pixel $\boldsymbol{p}_{c(i)}$ as correspondence:

$$\left(\boldsymbol{v}_i^{(t+1)}, \hat{\boldsymbol{v}}_{c(i)}^{(t)}\right)$$



Question: Are all pairs $\left(\boldsymbol{v}_i^{(t+1)}, \hat{\boldsymbol{v}}_{c(i)}^{(t)}\right)$ valid correspondences?



Criteria: A pair is only valid if

- their positions are close to each other : $\left\| \boldsymbol{v}_i^{(t+1)} - \hat{\boldsymbol{v}}_{c(i)}^{(t)} \right\|_2 \leq \delta_D$
- their normal directions are similar : $\left\langle \boldsymbol{n}_i^{(t+1)} \,\middle|\, \hat{\boldsymbol{n}}_{c(i)}^{(t)} \right\rangle \leq \delta_N$

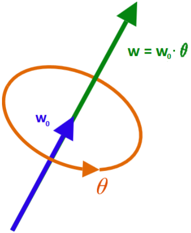
KinectFusion ICP

2. Find the optimal transformation $\boldsymbol{T}$ from

$$E_{icp} = \sum_i \left\langle \boldsymbol{T} \cdot \boldsymbol{v}_i^{(t+1)} - \hat{\boldsymbol{v}}_{c(i)}^{(t)} \,\middle|\, \hat{\boldsymbol{n}}_{c(i)}^{(t)} \right\rangle^2$$

In order to have a minimal representation for the rigid transformation, it is represented by a six-dimensional vector

$$\boldsymbol{\xi} = \begin{pmatrix} \boldsymbol{\omega}^T & \boldsymbol{t}^T \end{pmatrix}^T \in \mathbb{R}^6$$



This vector can be transformed to the original matrix by using the exponential mapping

$$\boldsymbol{T} = \tilde{\boldsymbol{\xi}} \quad , \quad \tilde{\boldsymbol{\xi}} = \begin{bmatrix} \exp\,[\boldsymbol{\omega}]_\times & \boldsymbol{t} \\ \boldsymbol{0}^T & 1 \end{bmatrix}$$

The exponential map effects a transformation from the axis-angle representation of rotations to rotation matrices. Given the vector $\boldsymbol{\omega}$ of the axis–angle representation, $[\boldsymbol{\omega}]_\times \in \mathbb{R}^{3\times 3}$ is its cross-product matrix, i.e. $\forall \boldsymbol{v} \in \mathbb{R}^3 : [\boldsymbol{\omega}]_\times \, \boldsymbol{v} = \boldsymbol{\omega} \times \boldsymbol{v}$. Then

$$\boldsymbol{R} = \exp\,[\boldsymbol{\omega}]_\times = \sum_{n=0}^{\infty} \frac{\left([\boldsymbol{\omega}]_\times\right)^n}{n!} = \sum_{n=0}^{\infty} \frac{\theta^n \left([\boldsymbol{\omega}_0]_\times\right)^n}{n!}$$

Note, that this definition implies the well known Rodrigues' formula

$$\boldsymbol{R} = \boldsymbol{I} + \sin(\theta)\,[\boldsymbol{\omega}_0]_\times + (1 - \cos(\theta))\left([\boldsymbol{\omega}_0]_\times\right)^2$$

Since only small rotations are assumed, the exponential mapping can be approximated linearly:

$$\boldsymbol{R} = \exp\,[\boldsymbol{\omega}]_\times = \sum_{n=0}^{\infty} \frac{\left([\boldsymbol{\omega}]_\times\right)^n}{n!} \approx \boldsymbol{I} + [\boldsymbol{\omega}]_\times$$

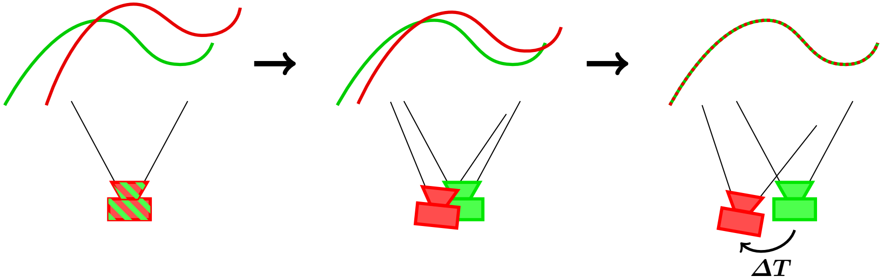
By plugging this into the error function and rearranging the terms, one gets the following result:

$$E_{icp} = \sum_i \left(\begin{bmatrix} \boldsymbol{v}_i^{(t+1)} \times \hat{\boldsymbol{n}}_{c(i)}^{(t)} \\ \hat{\boldsymbol{n}}_{c(i)}^{(t)} \end{bmatrix}^T \boldsymbol{\xi} + \left\langle \, \boldsymbol{v}_i^{(t+1)} - \hat{\boldsymbol{v}}_{c(i)}^{(t)} \, \middle| \, \hat{\boldsymbol{n}}_{c(i)}^{(t)} \, \right\rangle \right)^2 = \left\| \boldsymbol{J}_{icp}\, \boldsymbol{\xi} + \boldsymbol{r}_{icp} \right\|_2^2$$

This is a standard equation system that can be solved using the normal equation.

3. Finally, the estimated parameters $\boldsymbol{\xi}$ can be used to update the incremental transformation

$$\boldsymbol{\Delta T} \leftarrow \begin{bmatrix} \exp\left[\boldsymbol{\omega}\right]_\times & \boldsymbol{t} \\ \mathbf{0}^T & 1 \end{bmatrix} \cdot \boldsymbol{\Delta T}$$



These three steps are interated until convergence.

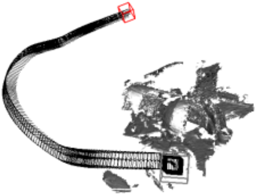
Advantages:

+ Intuitive formulation

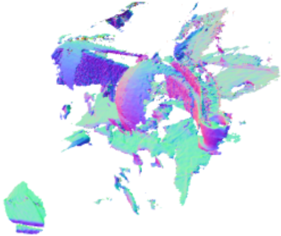
Disadvantages:

- Noise corrupts trajectory
- Too strong smoothing decreases accuracy

Reference



Trajectory



Reconstruction

(a) First RGB image $\boldsymbol{I}_{rgb}^{(t)}$

(b) Second RGB image $\boldsymbol{I}_{rgb}^{(t+1)}$

Observation: Depth images are noisy, but RGB images typically have much less noise

- Objects look very similar from the two view points
- Most objects reflect mainly diffuse $\rightarrow$ Assume Lambertian surfaces

Photometric ICP

Idea: Maximize Photo-Consistency $\Leftrightarrow$ Minimize the photometric error

$\boldsymbol{I}_{rgb}^{(t)}$

$\xrightarrow{\boldsymbol{\Delta T}}$

$\boldsymbol{\Delta T} \cdot \boldsymbol{I}_{rgb}^{(t)}$

$\boldsymbol{I}_{rgb}^{(t+1)}$

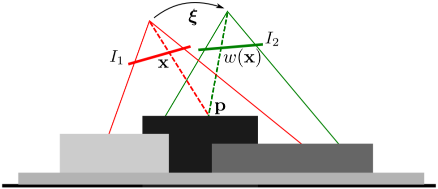
$$E_{rgbd} = \sum_i \left(\boldsymbol{I}_g^{(t)}(\boldsymbol{w}(\boldsymbol{\xi}, \boldsymbol{p}_i)) - \boldsymbol{I}_g^{(t+1)}(\boldsymbol{p}_i) \right)^2$$

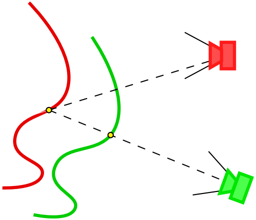
- $\boldsymbol{w}(\boldsymbol{\xi}, \boldsymbol{p}_i)$ is the warped pixel
- Only the intensity is used to increase robustness

The warped pixel $\boldsymbol{w}(\boldsymbol{\xi}, \boldsymbol{p}_i)$ can be seen as the corresponding pixel:

$$\boldsymbol{p}_{c(i)} = \boldsymbol{w}(\boldsymbol{\xi}, \boldsymbol{p}_i)$$

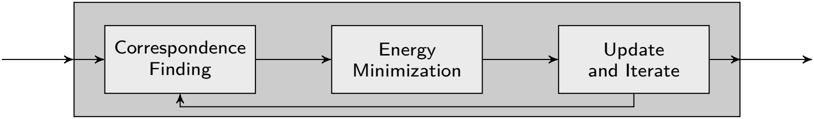
Thus, the warping operation in the Photometric ICP is closely related to the correspondence pairs of the KinectFusion ICP (in fact, they are the same).





This means that we can use the same correspondence criteria here plus an additional intensity-based one:

$$\left\|\boldsymbol{v}_i^{(t+1)} - \hat{\boldsymbol{v}}_{c(i)}^{(t)}\right\| \leq \delta_D \quad , \quad \left\langle \boldsymbol{n}_i^{(t+1)} \,\middle|\, \hat{\boldsymbol{n}}_{c(i)}^{(t)} \right\rangle \leq \delta_N \quad , \quad \left|\boldsymbol{I}_g^{(t+1)}(\boldsymbol{p}_i) - \boldsymbol{I}_g^{(t)}(\boldsymbol{p}_{c(i)})\right| \leq \delta_I$$



Now, the error function can be rewritten in the following way:

$$E_{rgbd} = \sum_i d(\boldsymbol{\xi}, \boldsymbol{p}_i)^2 \quad , \quad d(\boldsymbol{\xi}, \boldsymbol{p}_i) = \boldsymbol{I}_g^{(t)}(\boldsymbol{w}(\boldsymbol{\xi}, \boldsymbol{p}_i)) - \boldsymbol{I}_g^{(t+1)}(\boldsymbol{p}_i)$$

- Each error term can be linearly approximated by a Taylor expansion:

$$d(\boldsymbol{\xi}, \boldsymbol{p}_i) \approx d(\mathbf{0}, \boldsymbol{p}_i) + \frac{\partial d}{\partial \boldsymbol{\xi}}(\mathbf{0}, \boldsymbol{p}_i) \cdot \boldsymbol{\xi}$$

- Plugging this into the total error, we get the same representation as in the KinectFusion ICP:

$$E_{rgbd} \approx \left\| \boldsymbol{J}_{rgbd}\, \boldsymbol{\xi} + \boldsymbol{r}_{rgbd} \right\|^2$$

- After solving the normal equation, the estimate is updated:

$$\boldsymbol{\Delta T} \leftarrow \begin{bmatrix} \exp\,[\boldsymbol{\omega}]_\times & \boldsymbol{t} \\ \mathbf{0}^T & 1 \end{bmatrix} \cdot \boldsymbol{\Delta T}$$

Photometric ICP

Advantages:

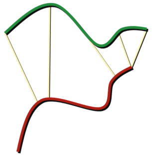
+ RGB images contain less noise

Disadvantages:

- Strongly depends on illumination

Question: How to compute $\boldsymbol{\Delta T}$?

KinectFusion ICP
(Newcombe et al., 2011)



Photometric ICP
(Steinbrücker et al., 2011)

Combined ICP
(Whelan et al., 2013)

So far, we have seen two different approaches to register RGB-D images. However, they can be brought to the same representation:

$$E_{icp} = \sum_i \left\langle \boldsymbol{T} \cdot \boldsymbol{v}_i^{(t+1)} - \hat{\boldsymbol{v}}_{c(i)}^{(t)} \,\middle|\, \hat{\boldsymbol{n}}_{c(i)}^{(t)} \right\rangle^2 \quad E_{rgbd} = \sum_i \left(\boldsymbol{I}_g^{(t)}(\boldsymbol{w}(\boldsymbol{\xi}, \boldsymbol{p}_i)) - \boldsymbol{I}_g^{(t+1)}(\boldsymbol{p}_i) \right)^2$$

$$\approx \left\| \boldsymbol{J}_{icp}\, \boldsymbol{\xi} + \boldsymbol{r}_{icp} \right\|^2 \qquad \approx \left\| \boldsymbol{J}_{rgbd}\, \boldsymbol{\xi} + \boldsymbol{r}_{rgbd} \right\|^2$$

Idea: Combine both approaches and define a new error function:

$$E_{combined} = E_{icp} + \lambda\, E_{rgbd}$$

- The correspondences are now much more constrained (position, normal orientation, itensity) and bad pairs are less likely
- Solving the combined error is done exactly as before

Improvement: A more intuitive formulation is given by

$$E_{combined} = (1 - \lambda)\, w_{icp}^2\, E_{icp} + \lambda\, w_{rgbd}^2\, E_{rgbd}$$

where w_{icp} and w_{rgbd} are weights to map both error terms to the same range. Now, λ works as an interpolation weight.

Results: fr1 desk2 scene from the TUM RGB-D benchmark dataset

- Many geometric and visual features, objects close to camera

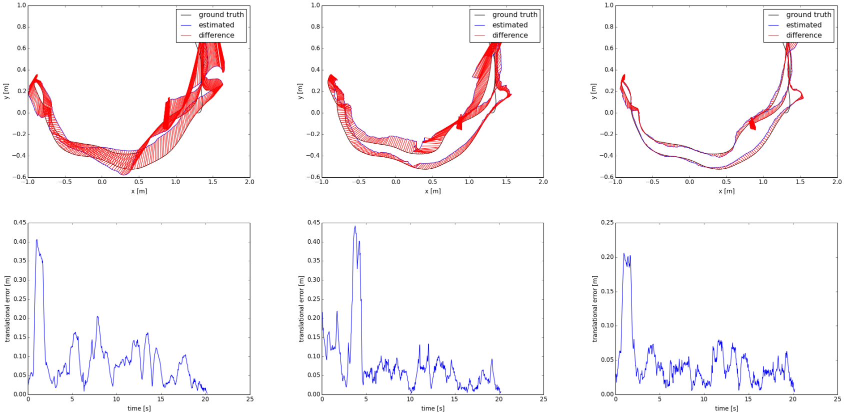


Figure: KinectFusion ICP, Photometric ICP, and Combined ICP with $\lambda = 0.9$

NDT

Another rigid registration method is the **N**ormal **D**istribution **T**ransform:[5]

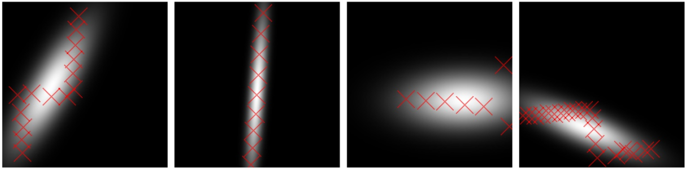
Idea: For each cell $\mathcal{C}_j$ in the 2D grid

- Consider all points of the point cloud $\mathcal{Q}$ that fall into this cell
- Compute the center of mass and estimate the covariance

$$\boldsymbol{\mu}_j = \frac{1}{K}\sum_{k=1}^{K}\boldsymbol{q}_k \quad , \quad \boldsymbol{\Sigma}_j = \frac{1}{K-1}\sum_{k=1}^{K}(\boldsymbol{q}_k - \boldsymbol{\mu}_j)(\boldsymbol{q}_k - \boldsymbol{\mu}_j)^T$$

- Define a probability function

$$p_{c(i)}(\boldsymbol{x}_i) \propto \exp\left(-\frac{1}{2}(\boldsymbol{x}_i - \boldsymbol{\mu}_{c(i)})^T \boldsymbol{\Sigma}_{c(i)}^{-1}(\boldsymbol{x}_i - \boldsymbol{\mu}_{c(i)})\right)$$



[5] P. Biber and W. Straßer. "The Normal Distributions Transform: A New Approach to Laser Scan Matching". In: *IROS*. IEEE, 2003, pp. 2743–2748.

Now the transformation $\boldsymbol{T} = (\boldsymbol{R}, \boldsymbol{t})$ can be found by

$$\arg\max_{\boldsymbol{T}} \sum_{i=1}^{n} \exp\left(-\frac{1}{2}\left(\boldsymbol{T}\,\boldsymbol{p}_i - \boldsymbol{\mu}_{c(i)}\right)^T \boldsymbol{\Sigma}_{c(i)}^{-1}\left(\boldsymbol{T}\,\boldsymbol{p}_i - \boldsymbol{\mu}_{c(i)}\right)\right)$$

A solution is found using the standard Newton method.

- Derivatives can be computed analytically

Advantages:

- No computation of point correspondences
- Very robust matching due to probabilistic formulation

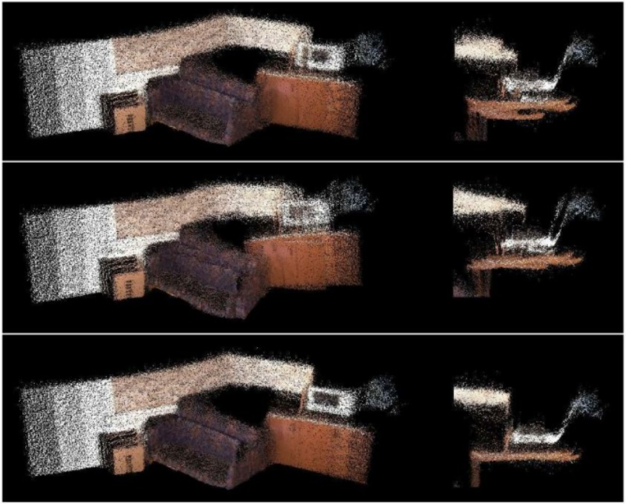
Extensions:

- Formulation in 3D[6] and incorporation of color information[7]

[6] M. Magnusson et al. "Scan Registration for Autonomous Mining Vehicles using 3D-NDT". In: *Journal of Field Robotics* 24.10 (2007), pp. 803–827.

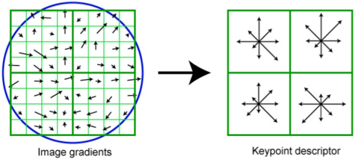
[7] B. Huhle et al. "Registration of Colored 3D Point Clouds with a kernel-based extension to the Normal Distributions Transform". In: *Robotics and Automation, 2008. ICRA 2008. IEEE International Conference on*. IEEE. 2008, pp. 4025–4030.

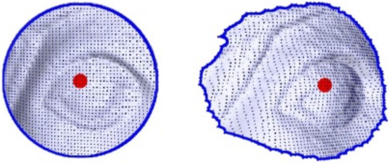
Results:

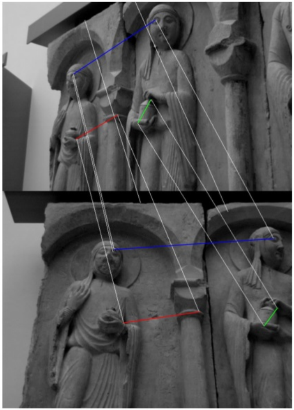


Idea: Align robust features[8][9]

1. Detect feature points in 2D image and 3D scan (SIFT, . . .)
2. Match the features (RANSAC)



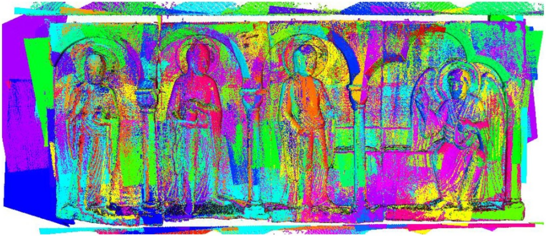




[8]G. H. Bendels et al. "Image-Based Registration of 3D-Range Data Using Feature Surface Elements". In: *VAST*. Citeseer. 2004, pp. 115–124.

[9]C. Wang and X. Guo. "Feature-based RGB-D camera pose optimization for real-time 3D reconstruction". In: *Computational Visual Media* 3.2 (2017), pp. 95–106.

Results:



Rendered Point Set

[1] G. H. Bendels, P. Degener, R. Wahl, M. Körtgen, and R. Klein. “Image-Based Registration of 3D-Range Data Using Feature Surface Elements”. In: *VAST*. Citeseer. 2004, pp. 115–124.

[2] P. Besl and N. D. McKay. “A Method for Registration of 3-D Shapes”. In: *Pattern Analysis and Machine Intelligence, IEEE Transactions on* 14.2 (1992), pp. 239–256.

[3] P. Biber and W. Straßer. “The Normal Distributions Transform: A New Approach to Laser Scan Matching”. In: *IROS*. IEEE, 2003, pp. 2743–2748.

[4] Y. Chen and G. Medioni. “Object Modeling by Registration of Multiple Range Images”. In: *Robotics and Automation, 1991. Proceedings., 1991 IEEE International Conference on*. IEEE. 1991, pp. 2724–2729.

[5] B. K. Horn, H. M. Hilden, and S. Negahdaripour. “Closed-form Solution of Absolute Orientation using Orthonormal Matrices”. In: *JOSA A* 5.7 (1988), pp. 1127–1135.

[6] B. Huhle, M. Magnusson, W. Straßer, and A. J. Lilienthal. “Registration of Colored 3D Point Clouds with a kernel-based extension to the Normal Distributions Transform”. In: *Robotics and Automation, 2008. ICRA 2008. IEEE International Conference on*. IEEE. 2008, pp. 4025–4030.

[7] M. Magnusson, A. Lilienthal, and T. Duckett. “Scan Registration for Autonomous Mining Vehicles using 3D-NDT”. In: *Journal of Field Robotics* 24.10 (2007), pp. 803–827.

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[11] F. Steinbrücker, J. Sturm, and D. Cremers. "Real-time visual odometry from dense RGB-D images". In: *Computer Vision Workshops (ICCV Workshops), 2011 IEEE International Conference on*. IEEE. 2011, pp. 719–722.

[12] C. Wang and X. Guo. "Feature-based RGB-D camera pose optimization for real-time 3D reconstruction". In: *Computational Visual Media* 3.2 (2017), pp. 95–106.

[13] T. Whelan, H. Johannsson, M. Kaess, J. J. Leonard, and J. McDonald. "Robust real-time visual odometry for dense RGB-D mapping". In: *Robotics and Automation (ICRA), 2013 IEEE International Conference on*. IEEE. 2013, pp. 5724–5731.